

TECH STACK & PRACTICAL USAGE

Divya Mary John

A detailed overview of tools, frameworks, and real-world applications used in development and learning.

1. Development Environment & Coding Tools

I primarily use Visual Studio Code as my development environment for writing, debugging, and organizing code efficiently. It provides flexibility through extensions such as Live Server for real-time previews and Prettier for consistent formatting. I also use Git and GitHub for version control, which allows me to track changes, manage project versions, and collaborate effectively. A concrete example of using these tools is during my React To-Do List project. I used Visual Studio Code to structure the project and debug issues, while Git helped me commit changes after implementing each feature such as adding or deleting tasks. GitHub allowed me to maintain a clean repository and track the evolution of the project. This setup helped me understand real-world development workflows.

2. Web Frameworks

I use React.js to build interactive and dynamic web applications. React allows me to break down user interfaces into reusable components and manage state effectively. In my To-Do List application, I created separate components for task input and task display. I used state management to dynamically update tasks when users interacted with the application. This helped me understand how frontend frameworks handle real-time updates and user interaction without reloading the page.

3. AI/ML Frameworks

I have explored TensorFlow and scikit-learn to understand machine learning fundamentals. These frameworks are useful for building and training models. For example, I used scikit-learn to build a simple classification model. I performed steps such as data preprocessing, training the model, and evaluating its performance. This hands-on experience helped me understand how machine learning workflows operate in practice.

4. Agentic AI Tools

I have experimented with tools like LangChain and AutoGPT to understand how AI agents work. These tools enable multi-step reasoning and automation. In one experiment, I used LangChain to process input through multiple steps to generate structured output. This helped me understand how complex tasks can be broken down into smaller steps and executed sequentially using AI.

5. LLM Platforms

I regularly use ChatGPT as a development and learning assistant. It helps me debug code, understand difficult concepts, and generate solutions. For instance, during my DBMS preparation, I used ChatGPT to understand normalization concepts like 2NF and 3NF through step-by-step explanations. This made it easier to apply these concepts while solving problems.

6. AI Tools

I use GitHub Copilot to improve coding efficiency by providing real-time code suggestions. It helps reduce syntax errors and speeds up development. While working on React components, Copilot suggested optimized ways to write event handlers and manage state. This improved both my coding speed and overall code quality.

7. Cloud Platforms

I have experience using Firebase for deploying applications and managing backend services. Firebase simplifies hosting and provides features like authentication and database services. In a small project, I used Firebase to deploy my web application and explored authentication features. This gave me practical exposure to cloud-based deployment.

8. Databases

I use MySQL for relational databases and MongoDB for NoSQL data storage. MySQL is useful for structured data and SQL queries, while MongoDB provides flexibility with JSON-like data. During my DBMS coursework, I used

MySQL to practice normalization, schema design, and SQL queries. I also used MongoDB to perform basic CRUD operations in small applications, helping me understand differences between relational and non-relational databases.